

Self-powered highly stretchable ferroelectret nanogenerator towards intelligent sports

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ABSTRACT

Ferroelectret nanogenerators (FENGs), recognized for their porous structures that facilitate charge retention, thereby creating giant electric dipoles and exhibiting remarkable piezoelectric properties, are utilized in the development of various flexible transducers. However, despite their flexibility, most developed ferroelectret nanogenerators lack adequate stretchability and satisfactory transverse piezoelectric properties, significantly inhibiting their widespread deployment in wearable or skin-mounted electronics. Here, we introduce a highly stretchable ferroelectret nanogenerator (HS-FENG) built from laser-induced graphene (LIG), Ecoflex and anhydrous ethanol, demonstrating exceptional flexibility and stretchability, along with longitudinal and transverse piezoelectric effects. The stretchability of HS-FENG can reach a record of 468 %, while the quasi-static piezoelectric coefficients d_{33} and d_{31} are approximately 120 pC/N and 70 pC/N, respectively. To our knowledge, this is the first demonstration of the developed FENG with remarkably high stretchability. Furthermore, leveraging the performance of the created HS-FENG, we construct a skin-mounted intelligent kinesiology tape capable of effectively monitoring motion signals from human muscles and joints, thereby offering a deeper understanding of movement for users across different levels of physical activity, from professional athletes to individuals undergoing rehabilitation. The development of intelligent kinesiology tape exemplifies the potential of HS-FENG technology in enhancing professional athletic training and personalized healthcare. It contributes to the advancement of inconspicuous skin-mounted biomechanical feedback systems and human-machine interfaces, marking progress in the field.

1. Introduction

As we delve deeper into the era of artificial intelligence (AI), Big Data, and the Internet of Things (IoT) [1,2], we are witnessing transformative changes across numerous facets of life, and sports is no exception [3,4]. This transition into the digital intelligence era has brought about an increasing demand for innovative intelligent sports sensors that can monitor real-time information about body movement and muscle activity status. Such developments are not only guiding sports training [5,6], but they also address people's health concerns related to sports [7,8]. Traditional sports bracelets, which primarily monitor pulse and heart rate as indirect feedback on sports status, have

become inadequate to meet the growing needs of users [9,10]. This has spurred the rapid development of new wearable electronic sensors that offer unique functionalities, high reliability, and precise monitoring [11–21]. However, some sensors require power from traditional energy storage devices (like batteries), which face challenges such as limited lifespan, hazardous chemicals, large size, rigid structure, and heat production [22–24]. Additionally, some sensors, despite their flexibility, demonstrate limited stretchability and twistability [25], which restricts their applicability to specific scenarios or exercise modes and hinders precise real-time exercise guidance. To tackle these challenges effectively, there is a need to develop a skin-mounted electronic sensor [26–29] that integrates selfpowered capabilities, excellent flexibility,

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stretchability, and twistability, along with continuous usability and broad applicability in various scenarios.

Ferroelectrets, porous materials that lack molecular dipoles but demonstrate piezoelectric properties, have garnered considerable attentions [30–33]. These materials possess numerous internal voids that, when charged with high electric fields, form artificial giant dipoles, thereby exhibiting piezoelectricity [34–36]. Exploiting the properties of ferroelectret materials, researchers have developed a variety of ferroelectret nanogenerators (FENGs) [37–39] using polymers such as polypropylene (PP) [40], polylactic acid (PLA) [41,42], and polytetrafluoroethylene (PTFE) [43]. These FENGs offer multiple benefits such as strong piezoelectricity, self-powered operation, high sensitivity, cost-effectiveness, high mechanical flexibility, and low acoustic impedance [19]. Due to these characteristics, they are increasingly employed in applications that convert mechanical and acoustic energy into electrical signals, and vice versa. For instance, researchers have created wearable devices such as flexible bracelets and insoles using FENGs, which interact with other devices through human touch and acoustic feedback [16,17,43]. These devices exhibit potential for use in human movement and disability rehabilitation fields [24,44,45]. However, harnessing the merits of FENG to develop self-powered wearable motion sensors or robotic flexible electronic skins (e-skins) that can fully adhere to human skin, matching its flexibility, softness, elasticity, and malleability, remains a challenge [46–48].

In this study, we present a skin-mounted motion sensor constructed from a flexible and highly stretchable ferroelectret nanogenerator (HS-FENG). HS-FENG is composed of Ecoflex and laser-induced graphene (LIG). Ecoflex is a biodegradable and compostable polymer known for its excellent biocompatibility and biodegradability [49], and it possesses high viscosity, flexibility, and superior stretchability [50–52]. The process of introducing anhydrous ethanol into the Ecoflex solution, followed by hot curing, results in the creation of artificial voids. When subjected to high-voltage charging, these voids form numerous giant dipoles within HS-FENG, enabling the conversion of mechanical energy from human motion into electrical signals through its piezoelectric effect. The high stretchability and piezoelectric performance of HS-FENG have been demonstrated through the characterization of its stretchability and measurement of its piezoelectric response in both longitudinal and transverse force modes. To verify the applicability and functionality of the HS-FENG as skin-mounted sports sensors or e-skins, we developed a wearable intelligent kinesiology tape based on HS-FENG. These tapes were applied to the joint muscles of sports coaches and trainees for demonstration purposes. Owing to HS-FENG's piezoelectric effect, the mechanical energy generated during the movement of joint muscles is precisely captured. By comparing the different signals produced by the same movements of sports coaches and trainees, coaches can provide more accurate and professional sports guidance. The skin-mounted intelligent kinematic tapes based on HS-FENG have shown high sensitivity and stability in capturing human motion signals across a range of activities, including competitive sports training (e.g., basketball), anaerobic training (e.g., bodybuilding), and aerobic training (e.g., long-distance walking). We anticipate that the advancement of HS-FENG will introduce new strategies in the development of skin-mounted wearable electronics and create opportunities for applying HS-FENG intelligent kinesiology tape in professional sports training and rehabilitation.

2. Results and discussion

In recent years, the field of sports science has undergone a notable transformation, fueled by rapid advancements in wearable technology. In this paper, the development of an intelligent kinesiology tape for athletes utilizing HS-FENG technology is presented. Traditional kinesiology tapes have gained popularity in the sports and fitness industry, being commonly used by athletes and fitness enthusiasts. These tapes are applied to muscles or joints to relax muscles, optimize muscle

contraction ability, and rectify imbalanced muscle movements. Here, we aim to develop intelligent kinesiology tapes that maintain similar appearance and mechanical properties while offering advanced motion monitoring capabilities. Fig. 1a shows the process of HS-FENG based intelligent kinesiology tape adhered to human muscles or joints to collect and analyze motion signals. In future, if integrated with wireless transmission modules and accompanied by a specialized mobile application, the HS-FENG can monitor the athlete's movement in real-time; the data captured is analyzed by the mobile app, providing coaches with effective and valuable insights into the athlete's performance. Fig. 1b shows the intelligent kinesiology tape was manufactured by encapsulating the HS-FENG between two layers of medical-grade polyurethane (PU) films, which provides both packaging and adhesive functions for the HS-FENG. It also visually compares traditional kinesiology tape with the HS-FENG based intelligent kinesiology tape, showing their similarities and the potential for the developed intelligent kinesiology tape to replace traditional options.

2.1. Structure design and fabrication

The HS-FENG demonstrated in this paper, characterized by its high flexibility and stretchability, employs a standard sandwiched conductor-insulator-conductor structure (Fig. 1c), with Fig. 1d depicting a schematic illustration of its state following stretching and twisting. LIGs are integrated into the HS-FENG as two surface conductive layers, eliminating the need for adhesives and additional conductors, thereby simplifying the manufacturing process and enhancing mechanical strength [53–55]. LIG is fabricated through laser ablation of a polyimide (PI) film with a thickness of 125 μm . Fig. 1e presents the scanning electron microscope (SEM) image of the LIG layer on the surface of HS-FENG. It can be seen that the LIG on the HS-FENG surface is uniform and dense, which helps to improve the surface conductivity and stretching resistance of HS-FENG. Ecoflex serves as an excellent choice for an intermediate insulating layer material in HS-FENG, owing to its superior flexibility and stretchability [56,57]. It also exhibits enhanced electronegativity and improved charge collection ability [57,58], along with a high level of compatibility with LIG [59,60]. The introduction of anhydrous ethanol into the Ecoflex solution generates voids of various sizes and shapes. These voids form once the ethanol evaporates during the heating and curing stages. Fig. 1f shows the SEM image of the cross-section of the interlayer, revealing internal voids with diameters ranging from 50 to 300 μm . A complete HS-FENG was formed by preparing two layers of LIG as the surface electrodes, porous Ecoflex as the intermediate insulator layer, and Ecoflex as an adhesive to bond the three layers together with no heterogeneous interfaces. Fig. 1g reveals the cross-section of HS-FENG under an optical microscope, where the sandwich structure of the conductor-insulator-conductor is shown. Fig. 1h and i represent the initial state of HS-FENG, and its physical form under stretching and twisting, respectively. When HS-FENG is stretched and twisted, the shape of the voids inside the interlayer flatten and distort accordingly. The fabrication process of the HS-FENG involves affixing a polyimide film to an aluminum plate, shaping it with an L-shaped LIG using a CO₂ laser cutter (Fig. 2a), and placing mixed solution of Ecoflex and anhydrous ethanol over the shaped surface (Fig. 2b). The excess solution is removed and the substance is cured at 135 °C for 5 min (Fig. 2c), and a large number of voids are formed internally (Fig. S1). Next, the cured upper layer is flipped (Fig. 2d), and the same procedure is applied to form a lower layer. This lower layer is then bonded to the cured upper layer using Ecoflex (Fig. 2e). After curing the two bonded layers at 135 °C for 5 min, the entire device is carefully peeled off using a blade (Fig. 2f). The final device has symmetrical L-shaped LIG electrodes on the top and bottom surfaces with numerous voids in the middle, forming engineered giant dipoles with a porosity of 40.85 % (Fig. 2g). Lastly, enameled wires are connected to the top and bottom layers of LIG using conductive carbon glue.

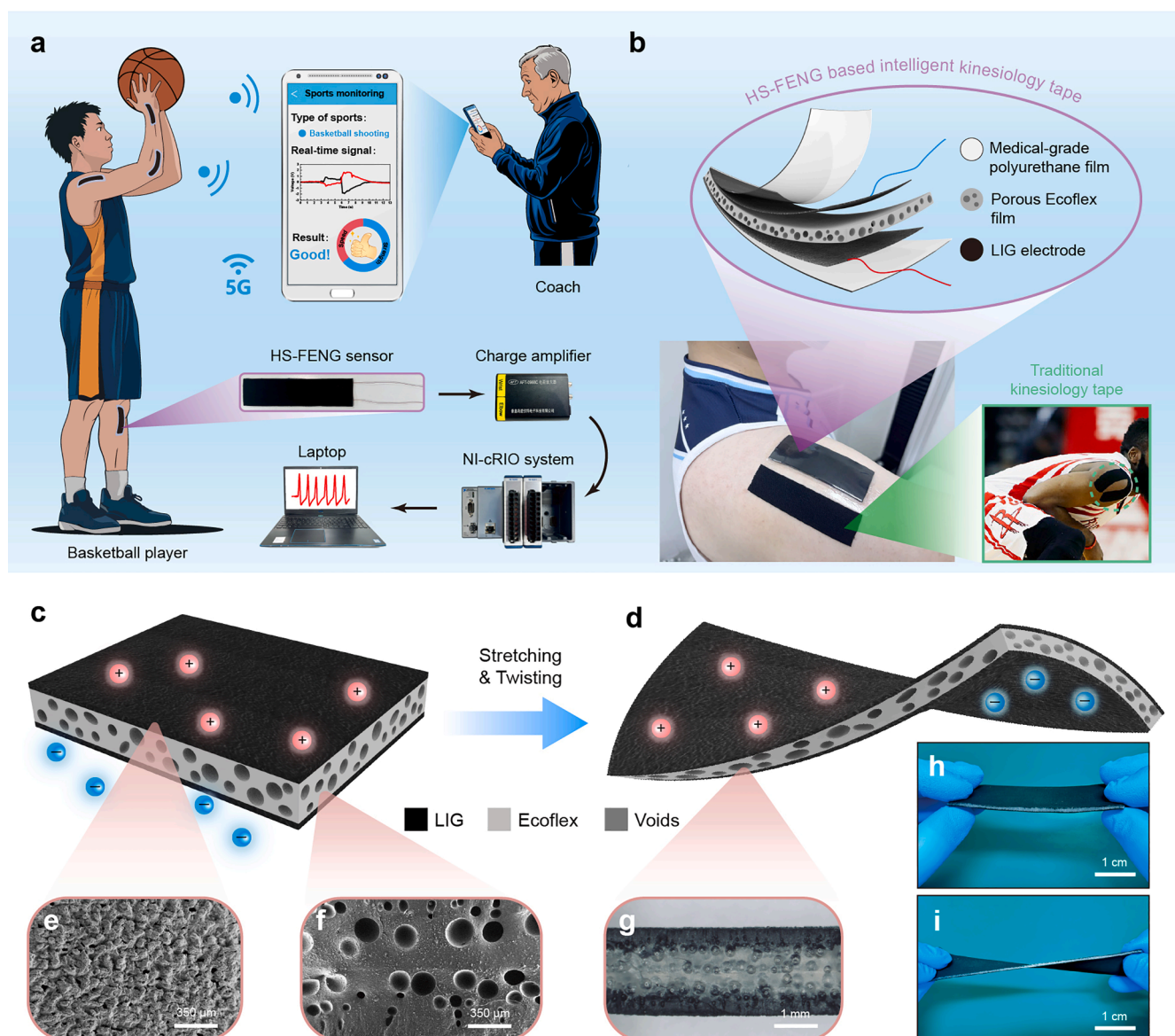


Fig. 1. Overview of the HS-FENG based intelligent kinesiology tape for sports. (a) Basketball players using intelligent kinesiology tapes to collect motion signals, and a coach wirelessly monitor and instruct players in the future. (b) The comparison between the traditional kinesiology tape and the intelligent kinesiology tape and the structure of the encapsulated intelligent kinesiology tape. Schematic diagrams depicting the HS-FENG in (c) relaxed initial state and (d) undergoing stretching and twisting deformations. (e) SEM image revealing the LIG surface morphology of HS-FENG. (f) SEM cross-sectional image showing the fabricated porous structure of HS-FENG. (g) Optical microscopy cross-sectional image showing the three-layer sandwiched architecture of HS-FENG. Photographs visually demonstrate HS-FENG in (h) relaxed initial state and (i) exhibiting stretchability and twistability under applied deformations.

2.2. Conductivity and stretchability of HS-FENG

To demonstrate the conductivity of the developed LIG under the enormous stretchability, an area of 40 mm × 40 mm HS-FENG was selected and anchored to test 8 points on its conductive LIG layer. These points were grouped in pairs to measure the resistance between them during the stretching process. Fig. 3a and b illustrate the tensile test photographs and the relationship between the average surface resistance of HS-FENG and strain when the direction of CO₂ laser scribing is parallel to the stretching direction, respectively. Fig. 3c and d show the tensile test photographs and the relationship between the average surface resistance of HS-FENG and strain when the direction of laser scribing is perpendicular to the stretching direction, respectively. It was revealed that regardless of whether the stretching direction and laser scribing direction are parallel or perpendicular, a consistent linear relationship is maintained between the measured resistance of HS-

FENG's electrode and the strain. Up to a strain of 100 %, no significant nonlinear phenomena were observed. Additionally, the electrical conductivity of HS-FENG remained consistently good during stretching (Fig. S2), providing operational and accuracy advantages for devices utilizing this technology. It is also noted that when the stretching direction of LIG is consistent with the engraving direction of the laser cutter, the conductivity of LIG is optimal. Unless otherwise stated, the stretching direction consistent with the laser scribing direction was chosen for subsequent experiments.

As a silicone rubber, Ecoflex offers not only flexibility but also remarkable stretchability. To investigate the stretchability of HS-FENG, tensile tests were conducted using a universal testing machine for both pure Ecoflex and HS-FENG, each with a thickness of 2 mm. Fig. 3e displays photographs of Ecoflex and HS-FENG in their initial and maximally stretched states, respectively. Fig. 3f shows the stress versus strain curve recorded by the universal testing machine during the stretching

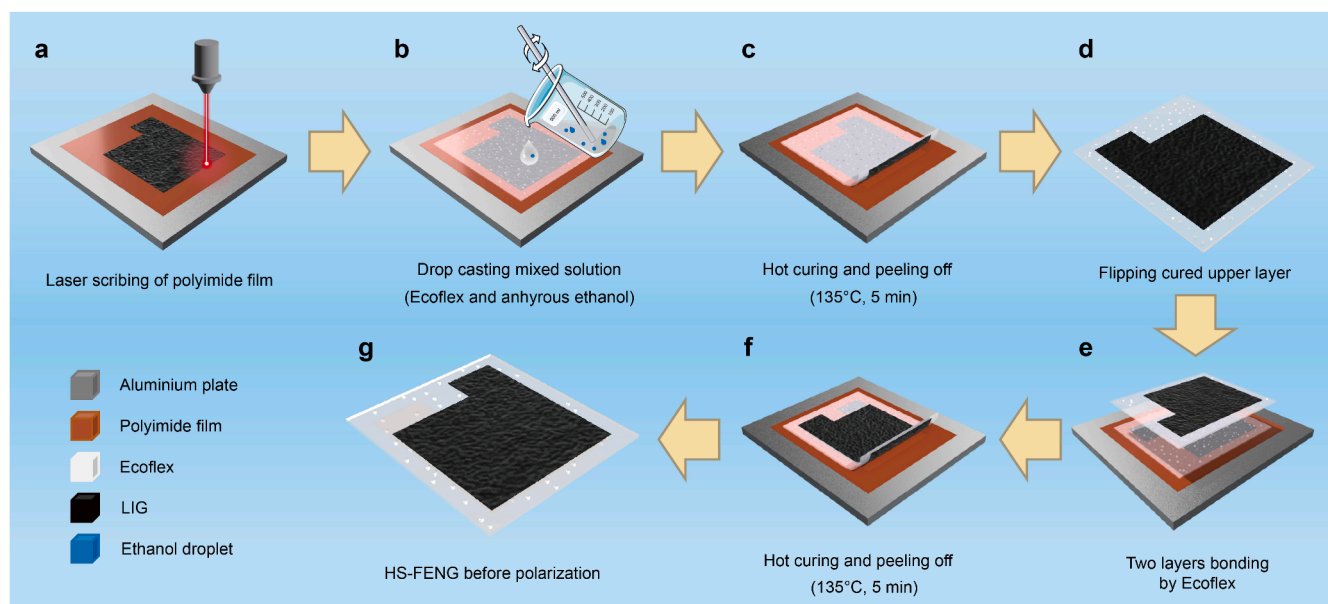


Fig. 2. Schematic diagram of the HS-FENG fabrication process. (a) CO₂ laser scribing of polyimide film to create LIG patterns. (b) Drop casting of vigorously mixed Ecoflex and anhydrous ethanol solution. (c) Hot curing and peeling off of the film from substrate. (d) Illustration of the flipped upper layer of HS-FENG. (e) Bonding of the cured upper layer to an identical uncured bottom layer. (f) Hot curing of the entire HS-FENG and peeling it off from substrate. (g) Illustration of the HS-FENG before polarization, showing its multi-layer void-containing structure.

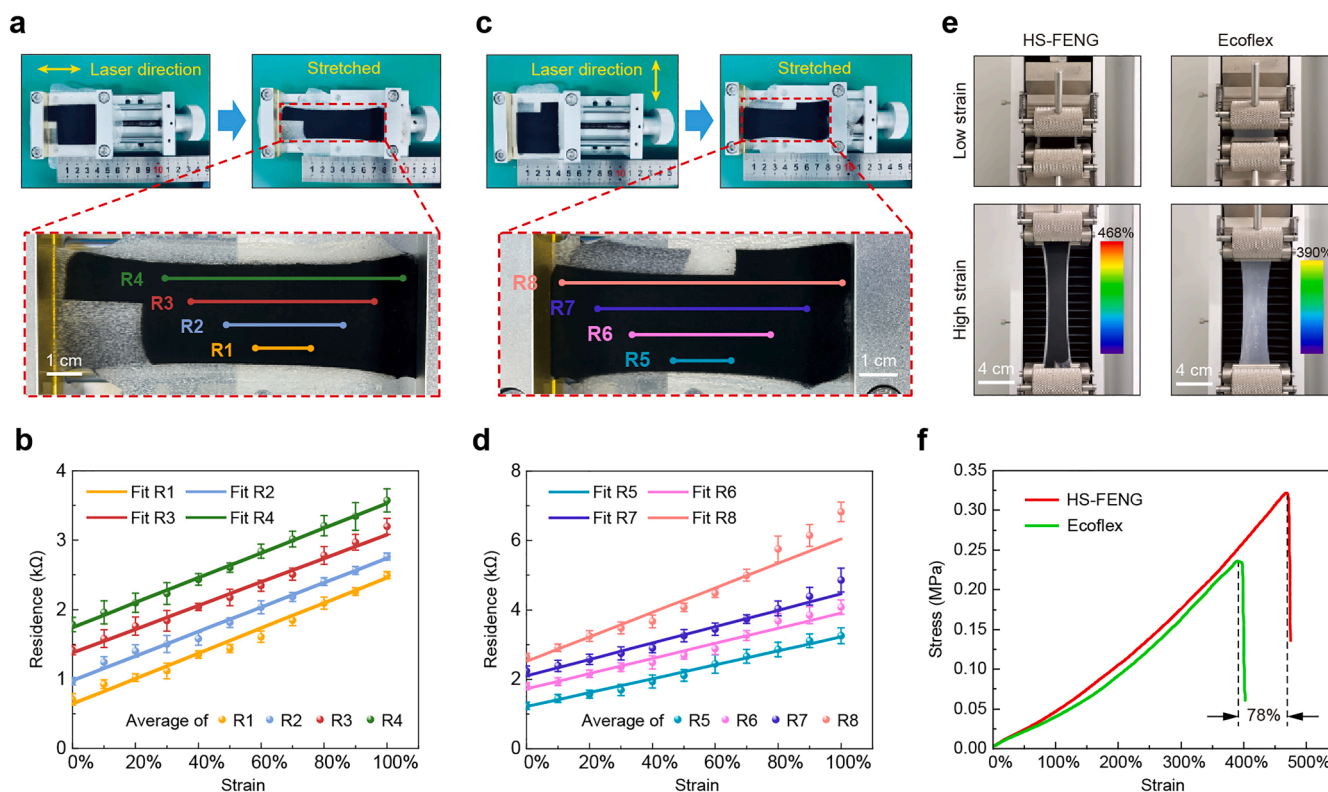


Fig. 3. Surface electrode conductivity and stretchability of the HS-FENG. (a) Photograph of tensile testing with the laser scribing direction parallel to the stretching direction. (b) Relationship between the average surface electrode resistance and strain when laser scribing is parallel to the stretching direction. (c) Photograph of tensile testing with the laser scribing direction perpendicular to the stretching direction. (d) Relationship between the average surface resistance and strain when laser scribing is perpendicular to the stretching direction. (e) Photographs showing the low and high strain states of the HS-FENG and Ecoflex before and after stretching. (f) Stress-strain curves of the HS-FENG and Ecoflex, showing their maximum stretchability limits.

process. The results demonstrate that HS-FENG exhibits superior stretching capabilities even compared to pure Ecoflex. Specifically, the elongation at break for pure Ecoflex is 390 %, whereas HS-FENG achieves 468 %. The further enhanced stretchability of HS-FENG can be

attributed to the presence of LIG on both the top and bottom surfaces. By controlling the laser fluence during scribing, the LIG grows like fuzz on the surface of the polyimide film, approximating a forest structure. When the Ecoflex solution is poured into the LIG, the solution flows into

the tiny gaps between all the LIG villi, completely saturating the entire LIG. Some of the longer LIG villi are deeply embedded in the Ecoflex (Fig. 1g). This LIG forest structure acts like numerous interlocking chains within the Ecoflex, thereby further increasing its overall stretchability [60,61].

2.3. Simulation analysis of HS-FENG

The HS-FENG, exhibiting both longitudinal and transverse piezoelectric effects, was modeled and qualitatively analyzed using the finite element method (FEM). Fig. 4a depicts a two-dimensional cross-sectional model of HS-FENG, comprising a middle layer with circular voids and LIG electrodes on the upper and lower surfaces. The model was constructed in COMSOL Multiphysics, where the surface charge densities of the upper and lower surfaces inside each void were set to -10^7 C/m² and 10^7 C/m², respectively, with the upper and lower electrodes grounded [40]. Fig. 4b, c illustrate the electric potential and electric field distributions across the HS-FENG cross-section in the absence of external force. It is evident that due to the charged surfaces within each void, an electric potential exists across the void surfaces and between them. Concerning the longitudinal piezoelectric effect, Fig. 4d, e show the simulated electric potential and electric field distributions when a vertical pressure is applied to the upper surface, causing a 0.6 mm downward displacement. The geometric shape of the voids, along with the electric potential and electric field distributions, are changed due to the external pressure. The size, shape, and density of the voids determine the vertical and horizontal stiffness of HS-FENG, which also influences its mechanical-electrical conversion capability. Regarding the transverse piezoelectric effect, Fig. 4f, g present the simulation results of HS-FENG under prescribed stretching. Notably, whether pressing or

stretching is applied, the internal circular voids transform into oval shapes. When both electrodes are grounded, the positive charge on the upper electrode and negative charge on the lower electrode simultaneously flow to the ground, resulting in a change in the accumulated charge density. By gradually increasing the applied pressure and stretch, and integrating the surface charge density of the electrodes, the relationships between the change in surface charge density versus the displacement due to pressing and stretching are obtained (Fig. 4h, i). It is observed that the variation in charge on the electrodes increases proportionally with the displacements in both circumstances. This macroscopic behavior of HS-FENG resembles that of intrinsic piezoelectric materials.

2.4. Optimization of corona charging for HS-FENG

Being a non-polar material distinct from its traditional counterparts that exhibit spontaneous polarization, HS-FENG forms internal giant dipoles through high-voltage point-to-plane corona discharge. This process creates giant dipoles that facilitate the conversion of mechanical energy into electrical energy, as depicted in Fig. 5a. When a strong electric field is applied to the device, Paschen breakdown occurs within the voids, causing the ionization and separation of charges in the gas. The electrically charged gaps change their thickness and, consequently, their dipole moments under stress. For example, compressing the HS-FENG leads to a reduction in dipole moments. This change in dipole moments can cause electrons to move from the negatively charged electrode to the positively charged electrode, resulting in a potential difference between the electrodes (under open-circuit condition) or a flow of charge from one electrode to the other (under short-circuit condition). The longitudinal d_{33} coefficient describes the piezoelectric

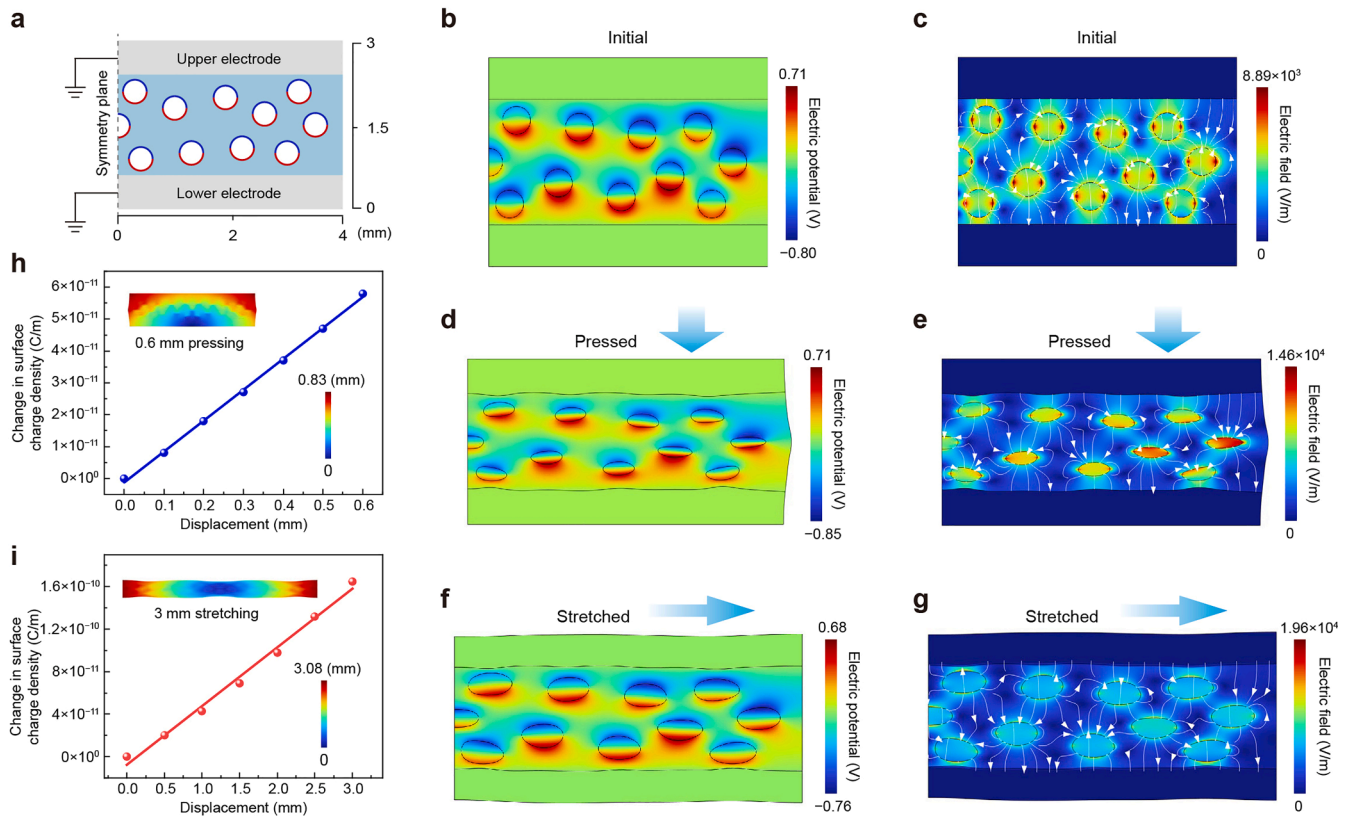


Fig. 4. Theoretical model and study of HS-FENG by FEM calculation. (a) Schematic model of a section of HS-FENG. (b) Electric potential distribution and (c) electric field distribution of the model with no external force. (d) Electric potential distribution and (e) electric field distribution of the model with a 0.6 mm downward displacement of the upper surface under a prescribed pressure. (f) Electric potential distribution and (g) electric field distribution of the model with a 1 mm displacement of right surface under a prescribed stretching. Relationship between the change in surface charge density and (h) pressing displacement, as well as (i) stretching displacement under external force. Insets show the geometries and displacement magnitudes of the model.

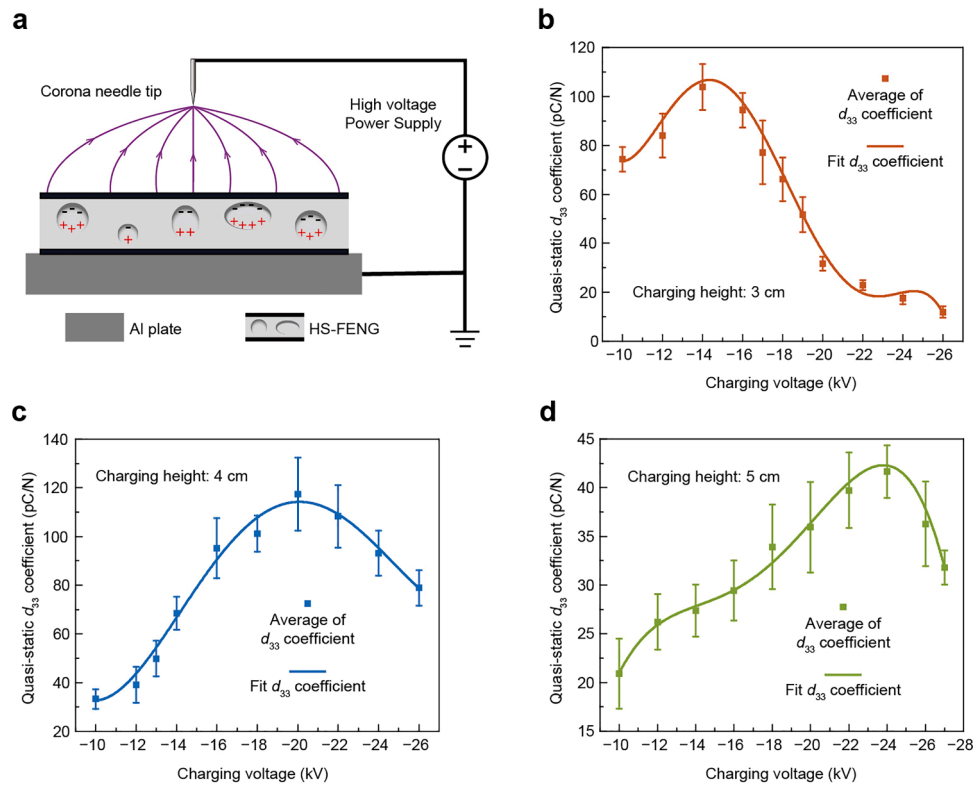


Fig. 5. Optimizing polarization to enhance piezoelectric performance of HS-FENG. (a) Schematic diagram depicting the corona charging process for fabricating HS-FENG. (b), (c), and (d) are the relationships between the quasi-static d_{33} coefficient and the applied high-voltage at the corona charging height of 3 cm, 4 cm and 5 cm, respectively.

response observed when an external mechanical force F is applied perpendicular to the HS-FENG surface along the polarization axis. To optimize the performance of HS-FENG, we studied how different conditions of corona discharge influence the d_{33} coefficient. In this case, we exposed the HS-FENG samples to corona discharge for 5 min at various corona charging heights (3 cm, 4 cm, 5 cm from corona needle tip to HS-FENG surface) and voltages (ranging from -10 kV to -26 kV) [62–64]. In the quasi-static longitudinal d_{33} coefficient measurement, a preload of $F_0 = 1$ N is applied to the entire electrode surface of the HS-FENG, followed by an application of a 3.5 N pressing force, which is then rapidly released (Fig. S3). The preload served to prevent the generation of interference signals when the applied force was released. Surface charge variations of HS-FENG were recorded using a charge amplifier and a data acquisition card. The quasi-static d_{33} coefficient is calculated as: $d_{33} = Q/F$, where Q denotes the amount of collected charge, and F denotes the amount of pressing force that is rapidly released. Fig. 5b–d illustrate the relationship between the quasi-static d_{33} coefficient and the charging voltage, varying the height of the corona needle tip. The results suggest that the longitudinal piezoelectric coefficient initially increases before decreasing as the charging voltage rises. The charging voltage at which the peak value of the quasi-static longitudinal d_{33} coefficient occurs is observed to increase with the corona charging height. According to the experiments, the maximum recorded value of the quasi-static longitudinal d_{33} coefficient is 120 pC/N, achieved at a corona charging height of 4 cm and a charging voltage of -20 kV, which was adopted as the optimal charging condition for the fabrication of HS-FENG.

2.5. Quasi-static and dynamic piezoelectricity: longitudinal and transverse modes

A precision infrared electrostatic field meter was utilized to measure the surface potentials on the upper and lower surfaces of the HS-FENG, which underwent a corona charging at a height of 4 cm for 5 min. Fig. 6a

depicts the effect of charging voltage on the upper and lower surface potential after polarization. The findings suggest that the potential of both the upper and lower surfaces reaches its maximum when the charging voltage is -20 kV, which corroborates the test results of the quasi-static d_{33} coefficient. Fig. 6b illustrates the quasi-static d_{33} coefficient of the HS-FENG subject to various pressure loads. It is evident that as load pressure increases, the quasi-static d_{33} coefficient initially escalates before progressively declining within the range of 50–130 pC/N. Notably, at a lower pressure range (0.6–0.9 kPa), the quasi-static d_{33} coefficient appears to stabilize, reaching its apex at approximately 130 pC/N. Upon conducting multiple tests, the piezoelectric coefficient demonstrated relative stability under diverse static and dynamic loads. The modulation of the d_{33} coefficient in response to applied pressure can be ascribed to the nonlinear correlation between Young's modulus and applied stress inherent in cellular polymer materials [65]. Fig. 6c displays the generated charge by HS-FENG under a static load ranging from 0 to 7 kPa. As static load increases, HS-FENG generates more charge, showing a positive correlation. However, due to its finite charge storage capacity, the rate of increase in generated charge gradually slows down, consistent with FENG's hysteresis effect [63].

Given the importance of the frequency dependence of the piezoelectric coefficient in applications such as wearable sports sensors, which often operate across a broad frequency range, we conducted tests on the longitudinal dynamic d_{33} coefficient of HS-FENG. This coefficient is calculated as $d_{33} = Q/(ma)$, where m represents the seismic mass applied to the surface of the HS-FENG, and a represents the dynamic acceleration of the seismic mass excited by a vibration exciter. A seismic mass of 100 g was affixed to the upper surface of HS-FENG, and the area of the sample under the applied mass was measured as 3.8 cm^2 (Fig. S4). Fig. 6d depicts the variation of the longitudinal dynamic d_{33} coefficient of the sample at different vibration frequencies. The three curves represent the longitudinal dynamic d_{33} coefficients generated at pressures of 0.7, 0.8, and 0.9 kPa, respectively. The acceleration of the

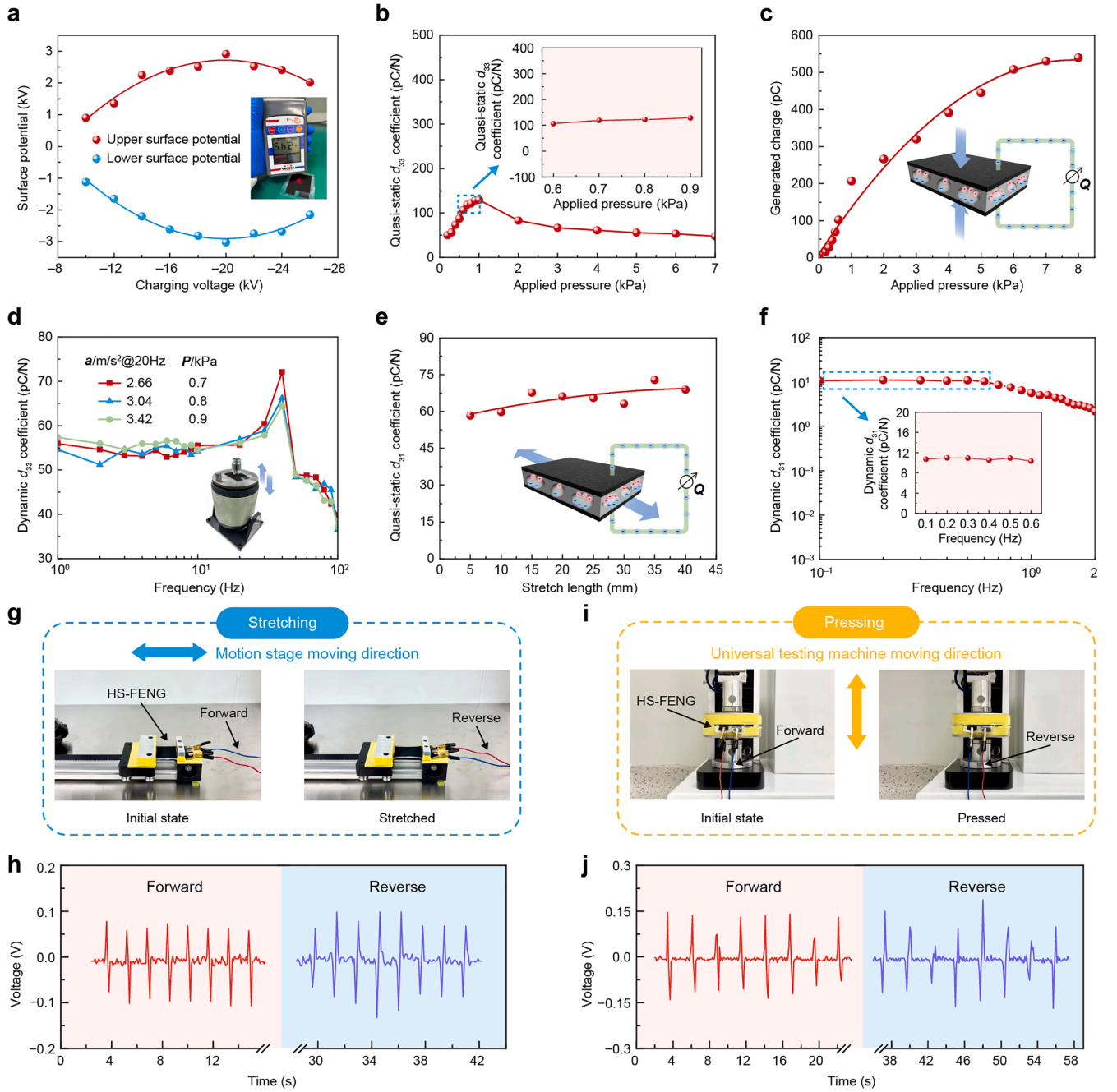


Fig. 6. Piezoelectricity measurements and Self-powered sensing measurement of the HS-FENG. (a) Effect of charging voltage on the surface potential of the HS-FENG. Inset shows the potential measurement. (b) Applied pressure dependence of the quasi-static d_{33} coefficient. Inset shows the values in the 0.6–0.9 kPa pressure range. (c) Generated charge on the HS-FENG electrodes at different applied pressures. Inset illustrates the d_{33} operating mode. (d) Frequency-dependent dynamic d_{33} coefficient under various accelerations with a seismic mass of 100 g. Inset shows the dynamic measurement with vibration exciter. (e) Quasi-static d_{31} coefficient of HS-FENG with an original length of 40 mm at different stretch lengths. Inset illustrates the d_{31} operating mode. (f) Frequency dependence of the dynamic d_{31} coefficient of the HS-FENG. Inset shows the values in the range of 0.1–0.6 Hz. (g) Photographs showing the repeated stretching test of the HS-FENG at its initial state with electrodes connected forward, and its stretched state with electrodes connected in reverse, using a stepper motor-driven motion stage. (h) Plot of the generated open-circuit voltage V_{oc} of the HS-FENG under stretching, before and after polarity switch. (i) Photographs showing the repeated pressing test of the HS-FENG at its initial state with electrodes connected forward, and its pressed state with electrodes connected in reverse, using a universal testing machine. (j) Plot of the generated open-circuit voltage V_{oc} of the HS-FENG under pressing, before and after polarity switch.

seismic mass is measured by an accelerator, and its magnitude is modulated by adjusting a power amplifier. It can be seen that the coefficients stabilize as the frequency increases within the low-frequency range up to approximately 20 Hz. Beyond this point, they experience a notable increase, reaching a peak around 40 Hz, before sharply declining at higher frequencies. It can be inferred that the resonance occurring at a vibration frequency of approximately 40 Hz leads to a

significant increase in the longitudinal dynamic d_{33} coefficient. Before reaching the resonance frequency, the coefficients of the sample at three different pressures remain stable. In the stable range, the longitudinal dynamic d_{33} coefficient of the sample reaches approximately 55 pC/N, which is around 46 % of the longitudinal quasi-static d_{33} coefficient. This difference can be attributed to the viscosity of Ecoflex as a silicone material [65].

Beyond studying traditional longitudinal piezoelectric responses, the transverse piezoelectric responses of HS-FENG are also investigated due to its unique feature of high stretchability. The d_{31} coefficient was measured to characterize the transverse piezoelectric effect of the HS-FENG (Fig. S5). During the test, the total width of the HS-FENG sample was represented as w , while the widths of the electrodes on both sides of HS-FENG were denoted by w_0 , and their length by l_0 . The width w of the sample is chosen to be slightly larger than the electrode width w_0 to prevent short-circuiting around the edges of the electrodes during significant stretching. When the HS-FENG reaches the preset length due to stretching by the universal testing machine, releasing the lower tensile grip manually separates one end of the sample, thereby releasing the tension. During this process, the charge produced by the HS-FENG is collected via a charge amplifier and then recorded by a data acquisition card. The quasi-static transverse coefficient d_{31} is given by $d_{31} = Qt/(F'l_0)$, where t represents the thickness of the HS-FENG, and F' is the force applied to the area with electrodes, expressed as [41]: $F' = Fw_0/w$. Here, F denotes the total force acting on the HS-FENG by the universal testing machine. Fig. 6e displays the quasi-static transverse d_{31} coefficient of HS-FENG with an original length of 40 mm at various stretch lengths. The quasi-static transverse d_{31} coefficient shows a slight but steady increase with the stretch length, within the range of 60–70 pC/N. This is a notably high d_{31} coefficient for a flexible FENG [66,67].

In addition to quasi-static measurements, dynamic measurements of the coefficient d_{31} of the HS-FENG were also conducted (Fig. S6). The frequency dependence of the dynamic d_{31} coefficient of the HS-FENG sample is presented in Fig. 6f. In this case, the area of HS-FENG under stretching has a size of 40 mm × 40 mm, with an additional 10 mm stretching controlled by a stepper motor-driven motion stage. The motion stage varied its stretching cycle frequency from 0.1 Hz to 2 Hz. The measured dynamic d_{31} coefficients ranged between 3 and 11 pC/N. The d_{31} coefficient initially stabilizes with increasing stretching cycle frequency, as depicted in the inset of Fig. 6f, and subsequently declines beyond approximately 0.6 Hz. This trend can be attributed to the viscosity of Ecoflex and the increase in the Young's modulus of the polymer film under higher excitation frequencies [68].

2.6. Self-powered sensing performance of HS-FENG

To evaluate the self-powered sensing performance of the HS-FENG, cyclic tensile tests using a stepper motor-driven motion stage to assess its transverse piezoelectric response were conducted. During these tests, the open-circuit voltage V_{oc} generated by the HS-FENG across its two electrodes was directly measured using the NI-cRIO system without the use of any charge amplification instruments. During signal acquisition, both ends of the HS-FENG were mounted on the motion stage and secured using preloading blocks that were wrapped with insulating tape (Fig. 6g). The voltage signal generated by the HS-FENG under horizontal tension with an exposed electrode area of 40 mm × 40 mm is depicted in Fig. 6h. Additionally, we utilized a universal testing machine to repeatedly apply pressure to the HS-FENG and then release it to characterize its longitudinal piezoelectric response (Fig. 6i). In this process, the HS-FENG was centrally positioned on the compression platens, which were wrapped with electrical tape for insulation, and V_{oc} was measured. Fig. 6j shows the voltage signal generated by the HS-FENG under vertical pressure with a force contact area of 40 mm × 40 mm. As observed, when stretched or pressed quickly, the magnitude of V_{oc} surges; conversely, upon release of the external force, the signal undergoes a similar change, but with an opposite sign. This behavior resembles that of piezoelectric devices, although the effect arises from artificial giant dipoles rather than natural molecular dipoles. When the polarity of the signal acquisition electrode is switched, the voltage signal produced by the HS-FENG reverses. This characteristic is indicative of piezoelectric devices, confirming that the measured output signal originates from HS-FENG rather than being caused by artifacts from the measuring instruments themselves (Movie S1, S2; Note S1, S2). Using a

stepper motor to cyclically stretch HS-FENG to a strain level of 50 %, and after 500 cycles of cyclic stretching, HS-FENG still returned to its original length after unloading the force and continued to be flexible and strongly stretchable.

2.7. Sports demonstrations of HS-FENG based intelligent kinesiology tape

HS-FENG may be poised as an excellent candidate for next-generation intelligent kinesiology tapes, characterized by its self-powered operation, superior flexibility and stretchability, high sensitivity, and cost-effectiveness. If necessary, HS-FENG can also function autonomously using the piezoelectric effect, enhancing its portability and making it suitable for diverse athletic settings. This feature is beneficial for athletes who are frequently on the move and may not have access to a power supply. Furthermore, the HS-FENG based intelligent kinesiology tape provides desirable flexibility, stretchability and stability, ensuring comfortable adherence to an athlete's skin during vigorous physical activities and the monitoring performance is not affected by temperature changes (Fig. S7). The quasi-static d_{33} coefficient remains stable between 20 °C and 40 °C (quasi-static d_{33} coefficient decreases by only 2 %). This adaptability makes it a versatile tool for athletes engaging in a wide range of sports and activities, as it can easily adjust to fit diverse body shapes and sizes.

The HS-FENG based intelligent kinesiology tape can accurately detect and track muscle and joint movements due to its effective longitudinal and transverse piezoelectricity, as well as its high flexibility and stretchability when applied in a skin-mounted manner. This feature is vital for athletes seeking to optimize their performance by identifying and rectifying their body movements. By providing real-time feedback on muscle movement patterns, the HS-FENG can aid athletes in refining their techniques and enhancing their overall performance. To demonstrate the capability of the HS-FENG based intelligent kinesiology tape in guiding athlete's movements, two such kinesiology tapes were utilized to monitor motion signals during competitive sports training. The tapes were worn—one on the right elbow and the other on the wrist—by two test subjects, including a basketball coach and a basketball amateur (Fig. 7a). These kinesiology tapes monitored the subjects' shooting actions at the free-throw line. Despite the movements of both subjects appearing similar, only the coach managed to score, while the amateur did not. The detected motion signals of the elbow and wrist of the subjects indicate that as the shooting action occurs, the force suddenly peaks due to muscle and joint contraction or extension, exerting force on the HS-FENG. The force subsequently decreases to a nearly constant value due to the HS-FENG's characteristics, akin to a piezoelectric device, which is more effective at detecting dynamic forces rather than static ones (Fig. 7b). The degree of muscle contraction or extension can be estimated from the magnitude of the signal peaks. Additionally, the direction of force change, i.e., whether the force acting on the kinesiology tape is increasing or decreasing, is represented by the positive or negative value of the output signal. The kinesiology tapes placed on the elbows and wrists of the two subjects had been pre-stretched at the initial position. During the shooting motion, the extension of the elbow caused the intelligent kinesiology tape skin-mounted at that joint to rapidly contract from its previously extended state; concurrently, the wrist flexion led to a rapid stretching of the tape skin-mounted at the wrist due to the exerted force (Movie S3, Note S3). The real-time, dynamic signal profiles captured by the intelligent kinesiology tapes accurately reflected the movements of the key muscles and joints involved in the shooting action (Fig. 7b). By analyzing these signal profiles, the coach could pinpoint the amateur's technique flaws and provide targeted guidance for improvement. For example, the amateur's signal exhibited an obviously lower amplitude compared to the coach's in this case, indicating inadequate elbow and wrist strength in the amateur, thereby highlighting the need for arm strength training.

In addition to competitive sports training, HS-FENG based intelligent kinesiology tapes also have a market potential in detecting muscle and

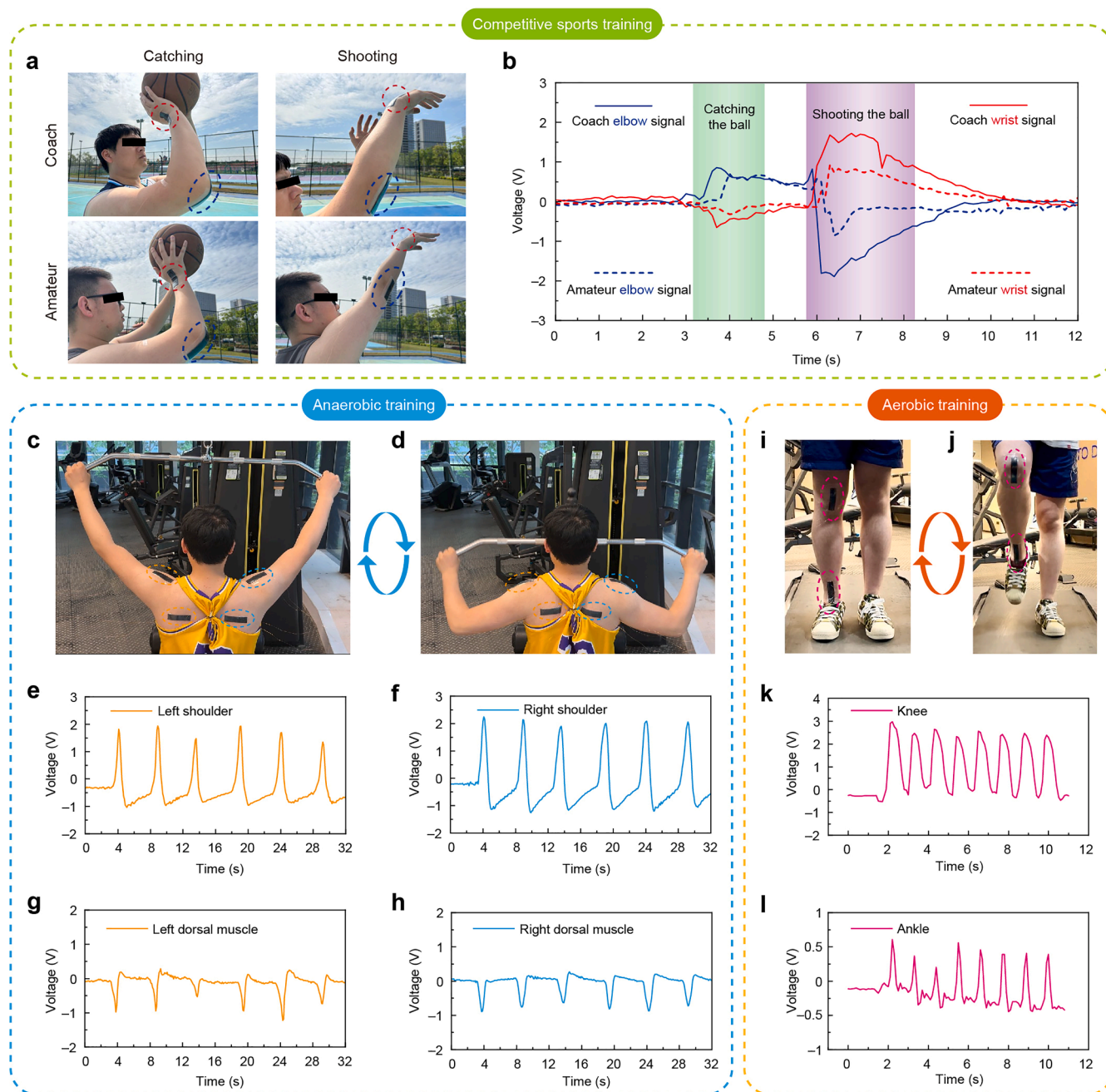


Fig. 7. Applications of HS-FENG based intelligent kinematic tapes in sports training. (a) Photographs showing the use of the intelligent kinematic tapes in competitive sports training, specifically for basketball shooting. (b) Real-time signal plots from the elbow and wrist of both a coach and an amateur during free throw training. (c, d) Photographs of using intelligent kinematic tapes in anaerobic training, specifically for lat pulldown exercise in bodybuilding. Signal plots acquired from the (e) left shoulder, (f) right shoulder, (g) left dorsal muscle, and (h) right dorsal muscle during the lat pulldown exercise. (i, j) Photographs of using intelligent kinematic tapes in aerobic training, specifically for long-distance walking on a treadmill. Signal plots generated from the (k) knee and (l) ankle by the intelligent kinematic tapes during the walking exercise.

joint motion signals during anaerobic and aerobic exercises performed by fitness professionals or enthusiasts. This can aid in establishing exercise standards and assessing training effectiveness, helping to track progress and identify potential flaws that may hinder an athlete's performance. As illustrated in Fig. 7c, d for anaerobic training, four HS-FENG based intelligent kinesiology tapes are skin-mounted on the left and right shoulders and back muscles of a fitness enthusiast during the use of the lat pulldown machine for bodybuilding (Movie S4, Note S4). This facilitates real-time recording of muscles' performance in the left and right shoulders (Fig. 7e, f), as well as the left and right back muscles (Fig. 7g, h) during shoulder and back workouts. By comparing the

recorded signals, insights could be gained into the fitness professionals' movement details, such as whether the targeted muscles were adequately stimulated for growth and the extent of that stimulation. Comparing collected muscle activity signals to standard signals also allows fitness professionals or enthusiasts to identify unconscious flaws like shoulder height discrepancies or imbalanced force exertion. This helps guide optimal exercise techniques and track progress effectively across different sets and repetitions.

When it comes to aerobic training, individuals engaged in endurance activities can assess their performances and receive recommendations to prevent injuries or expedite recovery by tracking essential body motion

signals. Fig. 7i, j illustrate the utilization of two HS-FENG based intelligent kinesiology tapes at the knee and ankle joints to monitor muscle and joint movement during aerobic exercises such as walking, running, and stair climbing. Fig. 7k, l depict the real-time motion signals captured by the HS-FENG based intelligent kinesiology tapes at the knees and ankles during the participant's treadmill usage (Movie S5, Note S5). This demonstration shows the potential of HS-FENG based intelligent kinesiology tapes in providing visual monitoring of rehabilitation progress for athletes recovering from injuries. For instance, by comparing the movement signals captured by the intelligent tapes before and after treatment, the effectiveness of the rehabilitation process can be assessed. This comparative analysis enables the customization of subsequent exercises to better align with the athlete's specific needs and recovery goals, thereby optimizing the rehabilitation journey.

Overall, the development of HS-FENG based intelligent kinesiology tape represents an endeavor in the field of wearable technology for sports and fitness. With its high flexibility, stretchability, twistability, sensitivity, and self-powered nature, it offers a versatile and user-friendly tool for athletes and sports enthusiasts alike. As the technology continues to evolve, it has the potential to improve the way we train and compete, helping athletes to realize their full potential and reach new heights in their respective sports. In addition to its applications in sports and fitness, the HS-FENG based intelligent kinesiology tape also holds promise for use in other fields. For instance, it could be used in rehabilitation settings to monitor the progress of patients undergoing physical therapy. By tracking their muscle status and providing feedback to therapists, these intelligent tapes may help patients receive the optimal treatment possible. As the landscape of personalized healthcare continues to evolve, HS-FENG could be at the forefront, actively contributing to innovative healthcare solutions. This aligns with the ongoing transition toward next-generation personalized healthcare, offering the potential to greatly enhance patient care and treatment outcomes.

3. Conclusion

To our knowledge, this is the first demonstration of a highly stretchable (468 % maximum strain) self-powered ferroelectric nanogenerator that has been developed by integrating Ecoflex, anhydrous ethanol, and LIG, showing exceptional flexibility, stretchability, twistability and strong longitudinal and transverse piezoelectric effects. These characteristics may forge new pathways for creating skin-mounted intelligent motion sensors and robotic e-skin. We conducted a series of experiments and performance tests to verify the HS-FENG's operational principles, mechanical properties, and self-powered sensing capabilities, thereby affirming its effectiveness and broad application potential. The fabrication process was refined by precisely controlling the laser fluence and the mass ratio of Ecoflex to anhydrous ethanol to optimize the formation of the LIG layers and the porous structure within the Ecoflex, and by optimizing the corona charging conditions to enhance performance. Experimental results proved that HS-FENG delivers good piezoelectric performance, with quasi-static piezoelectric coefficients d_{33} and d_{31} reaching 120 pC/N and 70 pC/N, respectively. Notably, HS-FENG can be stretched up to 468 %, vastly exceeding the stretchability of traditional FENG sensors, and even surpassing that of Ecoflex itself. Further testing under various pressures and vibration frequencies confirmed its reliability under both dynamic and quasi-static conditions. The practical application of the HS-FENG was demonstrated through the development of an intelligent kinesiology tape that accurately monitors realtime motion signals from human muscles and joints, offering valuable feedback and guidance to sports professionals and enthusiasts. This technology not only enhances posture correction, training evaluation, and injury recovery but also distinguishes itself with features such as high stretchability, twistability, flexibility, self-powering capability, cost-effectiveness, and ease of use. These attributes position the HS-FENG as a promising option for skin-

mounted wearable electronics, particularly suited to applications in professional sports training and personalized healthcare.

4. Experimental section

4.1. Fabrication of HS-FENG

A rectangular polyimide thin film with a thickness of 125 μm was secured on a 100 mm $\times$ 100 mm aluminum plate using four Kapton tape strips along the periphery. LIG conductive patterns were then scribed on the polyimide film's top surface using a laser cutter with a 50 W CO₂ laser source operating in rastering mode (4060, Ketai Inc.). Fig. S8 demonstrates various LIGs produced by varying the CO₂ laser cutter's cutting power, with the optimal power determined to be 11 W (Fig. S9). Next, anhydrous ethanol was vigorously stirred into an Ecoflex solution (Guoyuan Silicone Inc.) for 5 min, forming a uniform mixture. This Ecoflex/ethanol mixture was then uniformly spread over the LIG surface, covering it entirely, and excess mixture was removed by tilting the aluminum plate to allow it to flow away. The aluminum plate was then placed on a hotplate at 135 °C for 5 min to cure the mixture, after which the cured layer was carefully peeled off using a blade. The detached layer featured a LIG electrode on the top surface, a smooth bottom surface, and numerous voids in the middle. The same procedure was repeated to prepare another identical layer, but without curing. Subsequently, the first fabricated (cured) layer was combined with the second (uncured) layer by placing their LIG-free surfaces together, with the uncured Ecoflex serving as an adhesive between them. This combined two-layer film was then cured at 135 °C for 5 min and carefully peeled off using a blade, yielding the final structure with symmetrical L-shaped LIG electrodes on the top and bottom surfaces, enclosing numerous voids in the middle. During the polarization process, a high-voltage power supply (DW-P503-4ACD2, Dongwen Inc.) was used to generate a corona discharge between a corona needle tip and the surface of the HS-FENG. The distance between the tip and the HS-FENG surface (i.e., charging height) and the charging voltage was carefully regulated. The corona charging process involved applying high-voltage continuously for 5 min to ensure the creation of engineered dipoles within the voids of HS-FENG, which had a porosity of 40.85 %. The final step involved connecting enameled wires to the top and bottom layers of LIG using conductive carbon glue (16050, Ted Pella Inc.) and encapsulating the HS-FENG between two layers of medical-grade PU films (Hynaut Inc.). The role of the PU film package is to ensure that the HS-FENG is not affected by ambient humidity to the maximum extent.

4.2. Optimizing porous structure

Vigorously mixing anhydrous ethanol with Ecoflex solution resulted in the formation of abundant voids of varying sizes within the mixture. This phenomenon can be attributed to the high volatility of anhydrous ethanol. When the Ecoflex-ethanol mixture was heated to solidification, the anhydrous ethanol evaporated, leaving behind an array of voids within the solidified Ecoflex structure. The impact of different ethanol contents on the fractional void formation in Ecoflex was investigated (Table S1). The porosity of the Ecoflex was calculated using the formula: $P_{\text{porosity}} = 1 - \rho_{\text{cellular}}/\rho_{\text{solid}}$, where ρ_{solid} and ρ_{cellular} represent the densities of solid Ecoflex (1179 kg/m³) and the porous Ecoflex, respectively. As indicated in Supplementary Table 1, the highest porosity achieved was 40.85 %, corresponding to a mass ratio of 50:1 for Ecoflex to anhydrous ethanol. Fig. S10 provides a comparison of the porous Ecoflex structures formed according to different mass ratios. Since an increased number of internal voids in the FENG can enhance its charge storage capacity [35], the mass ratio of Ecoflex to anhydrous ethanol was chosen as 50:1 for the fabrication of the HS-FENG. Fig. S11 shows the average diameter size of the internal voids of HS-FENG is about 140 μm . Three samples were prepared according to the given optimal HS-FENG formulation and polarization recipe, showing consistent porosity and

piezoelectric properties, demonstrating that HS-FENG fabrication has good repeatability (Table S2).

4.3. Characterization and measurements

The surface morphology and the cross-section of the HS-FENG were characterized by field-emission scanning electron microscope (Ultra 55, Carl Zeiss Inc.) and stereo microscope (SMZ168-T, Motic Inc.). A precision infrared electrostatic tester (FMX-003, JSGONG Inc.) was utilized to measure the potential of the upper and lower surfaces of the HS-FENG at a corona charging height of 4 cm, after 5 min of corona charging. During the dynamic piezoelectric coefficient measurements of the HS-FENG, the seismic mass was shaken by a vibration exciter (VT-50, YMC Piezotronics Inc.), and the acceleration was measured by an accelerometer (22A25, YMC Piezotronics Inc.). For the sensing measurement of HS-FENG under stretching and pressing, a customized stepper motor-driven motion stage (Nema Inc.) was used to apply transverse deformation, while a universal testing machine (SSBC-500, Kezhun Test Instruments Inc.) was employed for longitudinal deformation. The generated signal was passed through a charge amplifier (AFT-0966C, BeiDaihe Inc.) and collected by a data acquisition card (DSO-2820, Virtins Inc.). To test the HS-FENG's self-powered sensing performance, the generated signals were acquired by an FPGA controller (CRIO-9075, National Instruments Inc.) with an analog input module (NI 9201, National Instruments Inc.). When monitoring the movement signals of human muscles and joints using the HS-FENG based intelligent kinesiology tape, the generated signals were passed through the charge amplifier, recorded by the FPGA controller, and demonstrated in real-time using the LabVIEW environment (National Instruments Inc.).

CRediT authorship contribution statement

Yiqin Wang: Writing – original draft, Validation, Investigation, Formal analysis, Data curation. **Xianfa Cai:** Writing – original draft, Validation, Investigation, Funding acquisition, Data curation. **Yufeng Guo:** Writing – review & editing, Project administration, Conceptualization. **Zhi Chen:** Investigation, Data curation. **Yunqi Cao:** Visualization, Investigation. **Wangdi Du:** Visualization, Investigation. **Tian Xia:** Writing – review & editing, Methodology. **Nelson Sepulveda:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Wei Li:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.nwnano.2024.100053](https://doi.org/10.1016/j.nwnano.2024.100053).

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